

Soil Health & Crop Recommendation Report

Report Date: November 16, 2025 at 14:04:22 UTC

Farmer Information	
Name:	test_farmer
Phone:	1234567890
Location:	Bangalore
Land Size:	5.50 acres

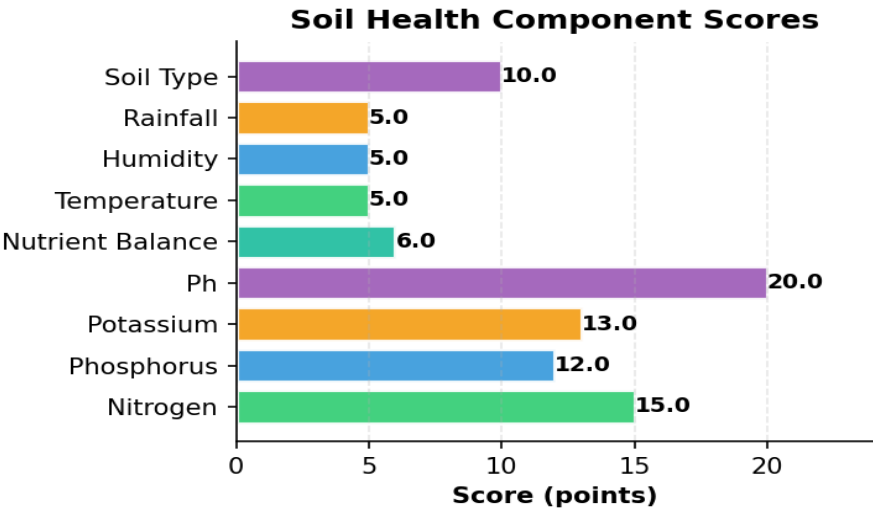
Parameter	Value	Unit
Nitrogen (N)	50.0	mg/kg
Phosphorus (P)	60.0	mg/kg
Potassium (K)	70.0	mg/kg
pH Level	6.50	
Temperature	25.5	°C
Humidity	65.0	%
Rainfall	120.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score	91.0/100	Excellent	
---------------------------	----------	-----------	---

Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	6.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	5.0	points

Soil Type	10.0	points
-----------	------	--------



Weather Suitability for Pigeonpeas

Overall Suitability Score: 7.3/10 ■ Good

Weather conditions are good for this crop

- ✓ Temperature 25.5°C is ideal for pigeonpeas
- ✓ Humidity 65.0% is ideal for pigeonpeas — reduces disease risk
- ■ Rainfall 120 mm is below requirement (600-1000 mm); irrigation strongly recommended

Recommended Crop

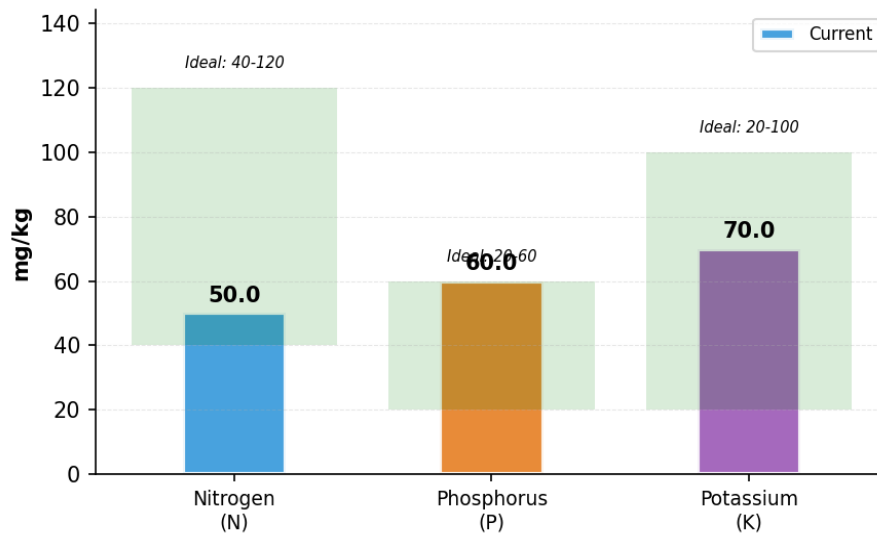
Predicted Crop	Pigeonpeas
Confidence Level	31.9%
Model Used	Random Forest

Top 3 Alternative Crop Options

Rank	Crop	Confidence (%)
1	Jute	14.7%
2	Chickpea	11.8%

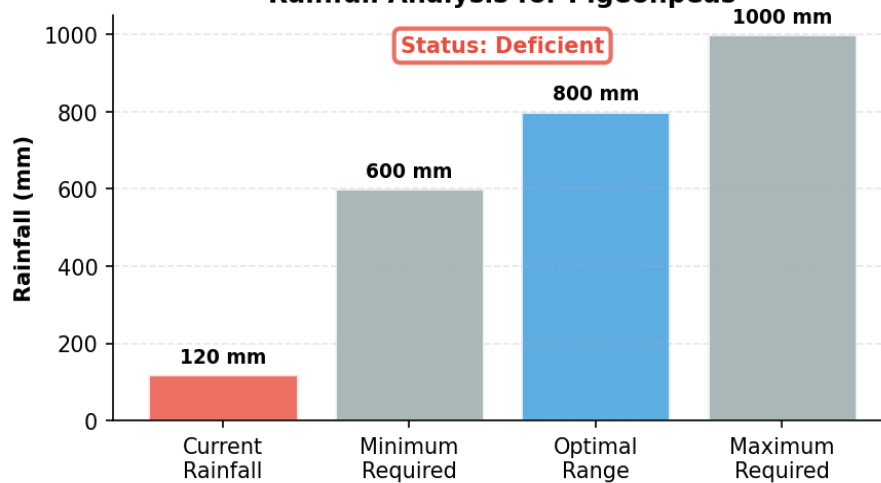
Soil NPK Levels Visualization

Soil NPK Levels vs Ideal Ranges



Rainfall Analysis

Rainfall Analysis for Pigeonpeas



Warnings & Observations

Action Required:

- Rainfall below optimal → Increase irrigation

Positive Indicators:

- ✓ Nitrogen level optimal
- ✓ Phosphorus level optimal
- ✓ Potassium level optimal
- ✓ pH optimal (6.5)
- ✓ Soil type: Alluvial Soil

■ Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for pigeonpeas.
- Nitrogen (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. Reduce nitrogen application to avoid excessive vegetative growth.
- Phosphorus (60.0 mg/kg) is within optimal range for pigeonpeas.
- Potassium (70.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 5.5 acres

Current Soil NPK: N=50.0 | P=60.0 | K=70.0 mg/kg

Optimal NPK for pigeonpeas: N=20 | P=50 | K=20 kg/acre

SOIL NUTRIENT STATUS

- ✓ Excellent: Soil NPK levels are already optimal for pigeonpeas

Maintain soil health through:

- Regular organic compost application
- Crop rotation practices
- Green manuring between seasons

ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 28-55 tons

- Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

- Application: 16-28 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

- Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local

agricultural extension recommendations for best results.

■ Crop-Specific Tips for Pigeonpeas

Drought-tolerant legume. Requires well-drained soil and warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

Soil Health & Crop Recommendation Report

Report Date: November 16, 2025 at 14:04:22 UTC

Farmer Information	
Name:	test_farmer
Phone:	1234567890
Location:	Bangalore
Land Size:	5.50 acres

Parameter	Value	Unit
Nitrogen (N)	50.0	mg/kg
Phosphorus (P)	60.0	mg/kg
Potassium (K)	70.0	mg/kg
pH Level	6.50	
Temperature	25.5	°C
Humidity	65.0	%
Rainfall	120.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score

91.0/100

Excellent

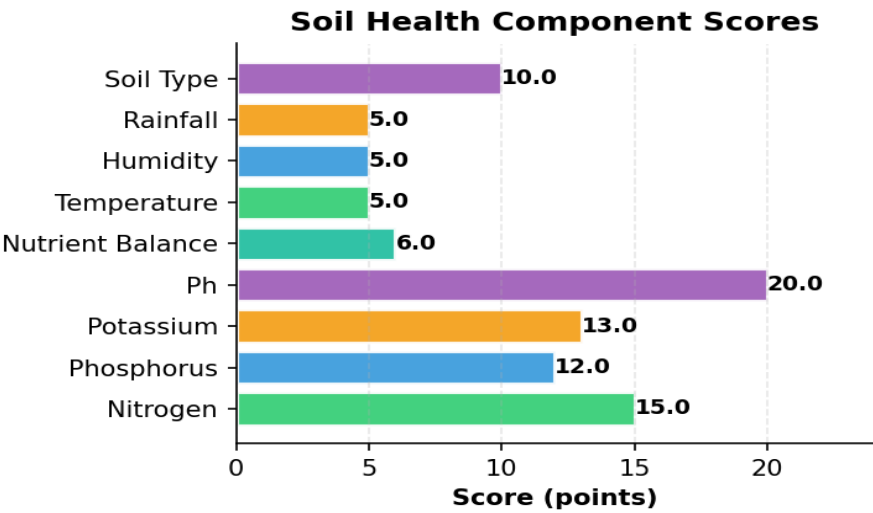


Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	6.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	5.0	points

Soil Health & Crop Recommendation System

Powered by ML + IoT + Weather Intelligence

Soil Type	10.0	points
-----------	------	--------



Weather Suitability for Pigeonpeas

Overall Suitability Score: 7.3/10 Good

Weather conditions are good for this crop

- ✓ Temperature 25.5°C is ideal for pigeonpeas
- ✓ Humidity 65.0% is ideal for pigeonpeas — reduces disease risk
- Rainfall 120 mm is below requirement (600-1000 mm); irrigation strongly recommended

Recommended Crop

Predicted Crop	Pigeonpeas
Confidence Level	31.9%
Model Used	Random Forest

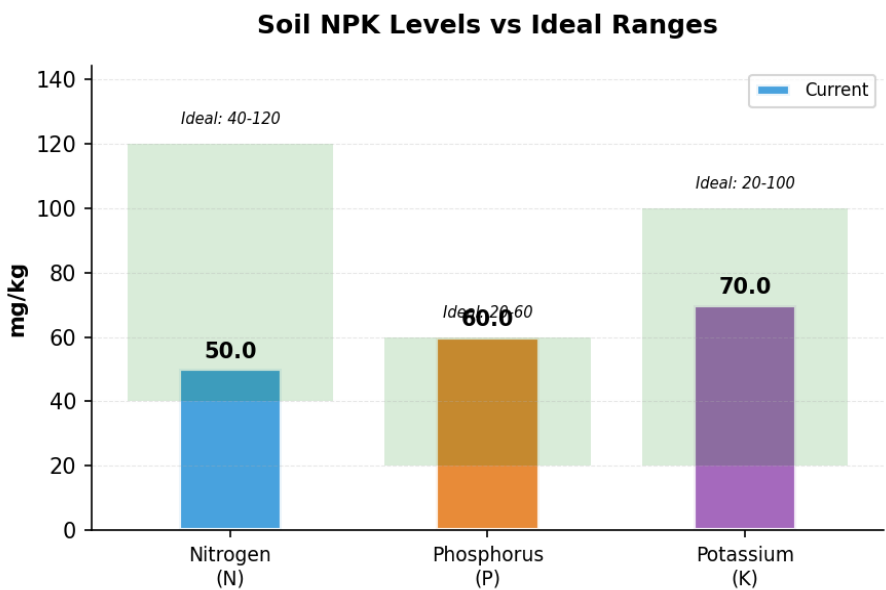
Top 3 Alternative Crop Options

Rank	Crop	Confidence (%)
1	Jute	14.7%
2	Chickpea	11.8%

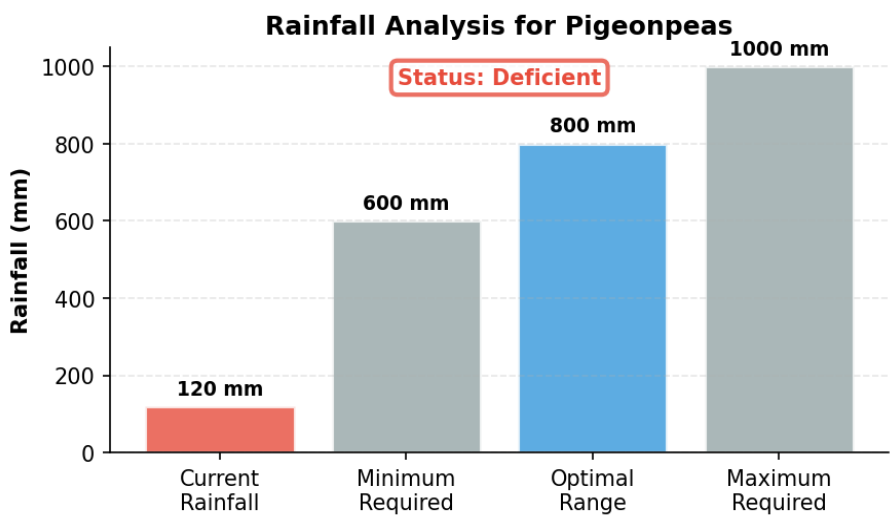
Soil NPK Levels Visualization

Soil Health & Crop Recommendation System

Powered by ML + IoT + Weather Intelligence



Rainfall Analysis



Warnings & Observations

Action Required:

- Rainfall below optimal → Increase irrigation

Positive Indicators:

- ✓ Nitrogen level optimal
- ✓ Phosphorus level optimal
- ✓ Potassium level optimal
- ✓ pH optimal (6.5)
- ✓ Soil type: Alluvial Soil

Soil Health & Crop Recommendation System

■ Powered by ML + IoT + Weather Intelligence

■ Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for pigeonpeas.
- Nitrogen (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. Reduce nitrogen application to avoid excessive vegetative growth.
- Phosphorus (60.0 mg/kg) is within optimal range for pigeonpeas.
- Potassium (70.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 5.5 acres

Current Soil NPK: N=50.0 | P=60.0 | K=70.0 mg/kg

Optimal NPK for pigeonpeas: N=20 | P=50 | K=20 kg/acre

SOIL NUTRIENT STATUS

✓ Excellent: Soil NPK levels are already optimal for pigeonpeas

Maintain soil health through:

- Regular organic compost application
- Crop rotation practices
- Green manuring between seasons

ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 28-55 tons

■■ Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

■■ Application: 16-28 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

■■ Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local

Soil Health & Crop Recommendation System

■ Powered by ML + IoT + Weather Intelligence

agricultural extension recommendations for best results.

■ Crop-Specific Tips for Pigeonpeas

Drought-tolerant legume. Requires well-drained soil and warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

Soil Health & Crop Recommendation System

■ Powered by ML + IoT + Weather Intelligence
